

Concept 3 | Rational Expressions

English Student Edition • Focused formulas and guided practice

Formula opener

Restriction

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Worked Solutions

Rational Expressions

Q001 Challenge

After simplifying this transfer chain before substitution, which result is correct for 2():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{2x(x+3)}{(x-3)(x-1)}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{2x(x+3)}{(x-3)(x-1)}$$

Worked Solutions

Rational Expressions

Q002 Challenge

After simplifying this transfer chain before substitution, which result is correct for 3():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3(x-3)(x+2)}{x(x+3)}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{3(x-3)(x+2)}{x(x+3)}$$

Worked Solutions

Rational Expressions

Q003 Challenge

After simplifying this transfer chain before substitution, which result is correct for 4():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8(x-4)(x+1)}{(x-3)(x-2)}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{8(x-4)(x+1)}{(x-3)(x-2)}$$

Worked Solutions

Rational Expressions

Q004 Challenge

After simplifying this transfer chain before substitution, which result is correct for 8():?

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8}{x}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{8}{x}$$

Worked Solutions

Rational Expressions

Q005 Challenge

After simplifying this transfer chain before substitution, which result is correct for 9():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$$

Worked Solutions

Rational Expressions

Q006 Challenge

After simplifying this transfer chain before substitution, which result is correct for 10():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$$

Worked Solutions

Rational Expressions

Q007 Written

Simplify $\frac{12ab^5}{9a^3b^2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{4b^3}{3a^2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{4b^3}{3a^2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{4b^3}{3a^2}$$

Q008 Written

Simplify $\frac{6x^2y^3}{20x^3y}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{3y^2}{10x}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{3y^2}{10x}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{3y^2}{10x}$$

Worked Solutions

Rational Expressions

Q009 Written

Simplify $\frac{6a^2bc^2}{8ab^4c^2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{3a}{4b^3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{3a}{4b^3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{3a}{4b^3}$$

Q010 Written

Simplify $\frac{2a^2b}{8a^3b^3}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{1}{4ab^2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{1}{4ab^2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{1}{4ab^2}$$

Worked Solutions

Rational Expressions

Q011 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{12ab^5}{9a^3b^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{4b^3}{a^2}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{4b^3}{a^2}$$

Q012 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6x^2y^3}{20x^3y}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9y^2}{10x}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9y^2}{10x}$$

Worked Solutions

Rational Expressions

Q013 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{6a^2bc^2}{8ab^4c^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3a}{b^3}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{3a}{b^3}$$

Q014 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{2a^2b}{8a^3b^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{1}{2ab^2}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{1}{2ab^2}$$

Worked Solutions

Rational Expressions

Q015 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{18a^3b^2}{24ab^5}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3a^2}{2b^3}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{3a^2}{2b^3}$$

Q016 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{-15x^4y^3}{20x^2y}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= -\frac{9x^2y^2}{4}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$-\frac{9x^2y^2}{4}$$

Worked Solutions

Rational Expressions

Q017 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{45a^2b^4c}{60a^5bc^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3b^3}{a^3c}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{3b^3}{a^3c}$$

Q018 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{14x^5y^2}{21x^2y^4}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{4x^3}{3y^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{4x^3}{3y^2}$$

Worked Solutions

Rational Expressions

Q019 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{32a^4}{48ab^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{2a^3}{b^2}$.
4. Retain the original denominator restrictions.

Correct option: A**Final answer**

$$\frac{2a^3}{b^2}$$

Q020 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{27x^3y^5}{36xy^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= 3x^2y^3$.
4. Retain the original denominator restrictions.

Correct option: B**Final answer**

$$3x^2y^3$$

Worked Solutions

Rational Expressions

Q021 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-42a^2c^3}{56a^5c}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= -\frac{3c^2}{2a^3}$.
4. Retain the original denominator restrictions.

Correct option: C**Final answer**

$$-\frac{3c^2}{2a^3}$$

Q022 Written

Simplify $\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}$.

Solution

1. Factor each polynomial: $\frac{a^3b^2}{3c}$ and $\frac{9c^3}{a^4b^2}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{3c^2}{a}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{3c^2}{a}$$

Worked Solutions

Rational Expressions

Q023 Written

Simplify $\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}$.

Solution

1. Factor each polynomial: $\frac{6bc^3}{5a^2}$ and $\frac{10a^4}{3b^3}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{4a^2c^3}{b^2}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{4a^2c^3}{b^2}$$

Q024 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{12c^2}{a}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{12c^2}{a}$$

Worked Solutions

Rational Expressions

Q025 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8a^2c^3}{b^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{8a^2c^3}{b^2}$$

Q026 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6a^2b}{5c} \times \frac{15c^2}{4ab^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{27ac}{2b^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{27ac}{2b^2}$$

Worked Solutions

Rational Expressions

Q027 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{9x^3y}{14a^2} \times \frac{21a^3}{6xy^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9ax^2}{y}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9ax^2}{y}$$

Q028 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-8ab^2}{3c^2} \times \frac{12c^3}{-5a^2b}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{64bc}{5a}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{64bc}{5a}$$

Worked Solutions

Rational Expressions

Q029 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{10x^2y^3}{9ab} \times \frac{27a^2b^2}{25xy}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{18abxy^2}{5}$.
4. Retain the original denominator restrictions.

Correct option: C**Final answer**

$$\frac{18abxy^2}{5}$$

Q030 Written

Simplify $\frac{x^2-4}{3x^2+7x+2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x-2}{3x+1}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x-2}{3x+1}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x-2}{3x+1}$$

Worked Solutions

Rational Expressions

Q031 Written

Simplify $\frac{2x^2-3x-5}{x^2+3x+2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{2x-5}{x+2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{2x-5}{x+2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{2x-5}{x+2}$$

Q032 Written

Simplify $\frac{x^2+3x}{x^2-9}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x}{x-3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x}{x-3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x}{x-3}$$

Worked Solutions

Rational Expressions

Q033 Written

Simplify $\frac{x^2+x-2}{x^2-5x+4}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x+2}{x-4}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x+2}{x-4}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x+2}{x-4}$$

Q034 Written

Simplify $\frac{2x^2-4xy+2y^2}{x^3+3x^2y-4xy^2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{2(x-y)}{x(x+4y)}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{2(x-y)}{x(x+4y)}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{2(x-y)}{x(x+4y)}$$

Worked Solutions

Rational Expressions

Q035 Written

Simplify $\frac{x^3-1}{x^2+2x-3}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x^2+x+1}{x+3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x^2+x+1}{x+3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x^2 + x + 1}{x + 3}$$

Q036 Written

Simplify $\frac{x^2-11x+24}{x^2-6x-16} \times \frac{x^2+2x}{x^2-6x+9}$.

Solution

1. Factor each polynomial: $\frac{(x-8)(x-3)}{(x-8)(x+2)}$ and $\frac{x(x+2)}{(x-3)^2}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{x}{x-3}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{x}{x-3}$$

Worked Solutions

Rational Expressions

Q037 Written

Simplify $\frac{x+1}{x^2-4} \times \frac{x^2-2x-8}{x^2+x}.$

Solution

1. Factor each polynomial: $\frac{x+1}{(x-2)(x+2)}$ and $\frac{(x-4)(x+2)}{x(x+1)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{x-4}{x(x-2)}.$
4. Retain the original denominator restrictions.

Final answer

$$\frac{x-4}{x(x-2)}$$

Q038 Written

Simplify $\frac{x^2-9}{x^2-2x} \times \frac{x^2-7x+10}{x^2+x-12}.$

Solution

1. Factor each polynomial: $\frac{(x-3)(x+3)}{x(x-2)}$ and $\frac{(x-5)(x-2)}{(x-3)(x+4)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{(x-5)(x+3)}{x(x+4)}.$
4. Retain the original denominator restrictions.

Final answer

$$\frac{(x-5)(x+3)}{x(x+4)}$$

Worked Solutions

Rational Expressions

Q039 Written

Simplify $\frac{x^2+x-6}{2x^2+x-6} \times \frac{x^2+6x+8}{3x^2-5x-2}$.

Solution

1. Factor each polynomial: $\frac{(x-2)(x+3)}{(x+2)(2x-3)}$ and $\frac{(x+2)(x+4)}{(x-2)(3x+1)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{(x+3)(x+4)}{(2x-3)(3x+1)}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$$

Q040 Written

Simplify $\frac{x^2-4x-5}{2x^2+5x+2} \div \frac{x^2-x-20}{2x^2-5x-3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-5)(x+1)}{(x+2)(2x+1)}$ and $\frac{(x-3)(2x+1)}{(x-5)(x+4)}$.
3. Cancel only common factors; the result is $= \frac{(x-3)(x+1)}{(x+2)(x+4)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{(x-3)(x+1)}{(x+2)(x+4)}$$

Worked Solutions

Rational Expressions

Q041 Written

Simplify $\frac{x^2-9}{x^2-x-2} \div \frac{2x^2+5x-3}{x+1}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-3)(x+3)}{(x-2)(x+1)}$ and $\frac{x+1}{(x+3)(2x-1)}$.
3. Cancel only common factors; the result is $= \frac{x-3}{(x-2)(2x-1)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{x-3}{(x-2)(2x-1)}$$

Q042 Written

Simplify $\frac{2xy}{3ab} \div \frac{8x^3y}{9a^2b^3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{2xy}{3ab}$ and $\frac{9a^2b^3}{8x^3y}$.
3. Cancel only common factors; the result is $= \frac{3ab^2}{4x^2}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{3ab^2}{4x^2}$$

Worked Solutions

Rational Expressions

Q043 Written

Simplify $\frac{2x^2-x-10}{x^2-2x-3} \div \frac{x^2+x-2}{x^2-6x+9}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x+2)(2x-5)}{(x-3)(x+1)}$ and $\frac{(x-3)^2}{(x-1)(x+2)}$.
3. Cancel only common factors; the result is $= \frac{(x-3)(2x-5)}{(x-1)(x+1)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{(x-3)(2x-5)}{(x-1)(x+1)}$$

Q044 Written

Simplify $\frac{x^2-3x-10}{2x^2+3x-2} \div \frac{x^2-2x-15}{6x-3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-5)(x+2)}{(x+2)(2x-1)}$ and $\frac{3(2x-1)}{(x-5)(x+3)}$.
3. Cancel only common factors; the result is $= \frac{3}{x+3}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{3}{x+3}$$

Worked Solutions

Rational Expressions

Q045 Written

Simplify $\frac{x^2-y^2}{x^2-2xy+y^2} \div \frac{x^2+xy}{x-y}.$

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-y)(x+y)}{(x-y)^2}$ and $\frac{x-y}{x(x+y)}.$
3. Cancel only common factors; the result is $= \frac{1}{x}.$
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{1}{x}$$

Q046 Written

Simplify $\frac{x^3+8}{x^2-x-6} \times \frac{x^2-6x+9}{x^2-2x+4} \div \frac{2x-6}{x^5-1}$ for $x = \sqrt{123}.$

Solution

1. Factor every polynomial and postpone substitution until the symbolic cancellation is complete.
2. The chain simplifies to $= \frac{(x-1)(x^4+x^3+x^2+x+1)}{2}.$
3. Substitute $x = \sqrt{123}$ only after simplifying: $= -\frac{1}{2} + \frac{15129\sqrt{123}}{2}.$
4. Check all original denominators and divisor-nonzero conditions.

Final answer

$$-\frac{1}{2} + \frac{15129\sqrt{123}}{2}$$

Worked Solutions

Rational Expressions

Q047 Challenge

Simplify

$\left(\frac{x+1}{x-2} - \frac{2}{x+3} + \frac{3}{x+3} - \frac{1}{x-2}\right)$; state every excluded value.

Solution

1. Group like denominators, then combine over the two-factor LCM.
2. The reduced expression is $= \frac{x^2 + 4x - 2}{(x-2)(x+3)}$.
3. The original denominators still control the domain.

Final answer

$$\frac{x^2 + 4x - 2}{(x-2)(x+3)}$$

Worked Solutions

Rational Expressions

Q048 Written

Simplify

$$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= x - 1$.
4. Retain all values excluded by the original denominators.

Final answer

$$x - 1$$

Worked Solutions

Rational Expressions

Q049 MCQ

Simplify

$$\frac{x^2 + 2x + 3}{x + 1} + \frac{2x - 1}{x + 1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x^2 + 4x + 2}{x + 1}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{x^2 + 4x + 2}{x + 1}$$

Worked Solutions

Rational Expressions

Q050 MCQ

Simplify

$$\frac{2x^2 - 5}{x^2 - 4} + \frac{x + 6}{x^2 - 4}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^2 + x + 1}{(x - 2)(x + 2)}$.
5. Keep every value excluded by an original denominator.

Correct option: D**Final answer**

$$\frac{2x^2 + x + 1}{(x - 2)(x + 2)}$$

Worked Solutions

Rational Expressions

Q051 MCQ

Simplify

$$\frac{3x^2 + x - 2}{(x - 2)(x + 1)} + \frac{4x - 3}{(x - 2)(x + 1)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$$

Worked Solutions

Rational Expressions

Q052 MCQ

Simplify

$$\frac{x^3 - 1}{x + 2} + \frac{2x^2 + x + 4}{x + 2};$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x^3 + 2x^2 + x + 3}{x + 2}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{x^3 + 2x^2 + x + 3}{x + 2}$$

Worked Solutions

Rational Expressions

Q053 Written

Simplify

$$\frac{3}{x^2 + x} - \frac{-x + 3}{x^2 + x}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{1}{x + 1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{1}{x + 1}$$

Worked Solutions

Rational Expressions

Q054 MCQ

Simplify

$$\frac{x^2 + 4x + 4}{x + 2} - \frac{x + 4}{x + 2}.$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x(x + 3)}{x + 2}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{x(x + 3)}{x + 2}$$

Worked Solutions

Rational Expressions

Q055 MCQ

Simplify

$$\frac{3x^2 - 2x + 1}{x^2 - 1} - \frac{2x - 5}{x^2 - 1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$$

Worked Solutions

Rational Expressions

Q056 MCQ

Simplify

$$\frac{x^2 - 6x + 8}{(x - 1)(x + 2)} - \frac{x^2 - 3x + 2}{(x - 1)(x + 2)}.$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= -\frac{3(x - 2)}{(x - 1)(x + 2)}.$
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$-\frac{3(x - 2)}{(x - 1)(x + 2)}$$

Worked Solutions

Rational Expressions

Q057 MCQ

Simplify

$$\frac{2x^3 + x}{x - 3} - \frac{x^2 + 7}{x - 3}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^3 - x^2 + x - 7}{x - 3}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{2x^3 - x^2 + x - 7}{x - 3}$$

Worked Solutions

Rational Expressions

Q058 Written

Simplify

$$\frac{x+4}{x^2-2x} - \frac{3}{x^2-3x+2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x+2}{x(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x+2}{x(x-1)}$$

Q059 Written**Simplify**

$$\frac{x-1}{x^2+4x+3} + \frac{2}{x^2+5x+6}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x}{(x+1)(x+2)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x}{(x+1)(x+2)}$$

Worked Solutions

Rational Expressions

Q060 MCQ

Simplify

$$\frac{x-1}{(x+1)(x+3)} + \frac{2x+1}{(x+3)(x+5)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$$

Worked Solutions

Rational Expressions

Q061 MCQ

Simplify

$$\frac{x+3}{(x+2)(x+4)} + \frac{x-2}{(x+4)(x+6)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$$

Worked Solutions

Rational Expressions

Q062 MCQ

Simplify

$$\frac{3x + 2}{(x + 1)(x + 4)} + \frac{x + 5}{(x + 4)(x + 7)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$.
5. Keep every value excluded by an original denominator.

Correct option: D**Final answer**

$$\frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$$

Worked Solutions

Rational Expressions

Q063 MCQ

Simplify

$$\frac{x+1}{(x+2)(x+5)} + \frac{4x-3}{(x+5)(x+8)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$$

Worked Solutions

Rational Expressions

Q064 MCQ

Simplify

$$\frac{2x + 1}{(x + 3)(x + 6)} + \frac{x + 4}{(x + 6)(x + 9)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$$

Worked Solutions

Rational Expressions

Q065 MCQ

Simplify

$$\frac{x-2}{(x+1)(x+5)} + \frac{3x+1}{(x+5)(x+8)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$$

Worked Solutions

Rational Expressions

Q066 Written

Simplify

$$\frac{x}{x-1} - \frac{2}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x+2}{x+1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x+2}{x+1}$$

Q067 Written**Simplify**

$$\frac{x+2}{x} - \frac{x-2}{x-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3x-2}{x(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3x-2}{x(x-1)}$$

Worked Solutions

Rational Expressions

Q068 Written

Simplify

$$\frac{2}{x-1} + \frac{3}{x^2+5x-6}$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x+15}{(x-1)(x+6)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x+15}{(x-1)(x+6)}$$

Worked Solutions

Rational Expressions

Q069 MCQ

Simplify

$$\frac{x+2}{(x+1)(x+3)} - \frac{2}{x+1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= -\frac{x+4}{(x+1)(x+3)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$-\frac{x+4}{(x+1)(x+3)}$$

Worked Solutions

Rational Expressions

Q070 **Written****Simplify**

$$\frac{1}{x+2} + \frac{1}{x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x}{(x-2)(x+2)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x}{(x-2)(x+2)}$$

Q071 Written**Simplify**

$$\frac{x+4}{x^2+2x} - \frac{x+5}{x^2+x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= -\frac{2}{x(x-1)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$-\frac{2}{x(x-1)}$$

Worked Solutions

Rational Expressions

Q072 Written

Simplify

$$\frac{x+8}{x^2+x-2} - \frac{x+5}{x^2-1}$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2}{(x+1)(x+2)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2}{(x+1)(x+2)}$$

Q073 Written**Simplify**

$$\frac{x-3}{x^2+x} + \frac{x}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x^2 - 4x + 3}{x(x-1)(x+1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x^2 - 4x + 3}{x(x-1)(x+1)}$$

Worked Solutions

Rational Expressions

Q074 **Written****Simplify**

$$\frac{4x}{x^2 - 1} - \frac{x - 1}{x^2 + x}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3x - 1}{x(x - 1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3x - 1}{x(x - 1)}$$

Q075 Written**Simplify**

$$\frac{1}{x^2 + 5x + 6} + \frac{1}{x^2 + 7x + 12}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2}{(x + 2)(x + 4)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2}{(x + 2)(x + 4)}$$

Worked Solutions

Rational Expressions

Q076 Written

Simplify

$$\frac{x-1}{x^2+3x+2} - \frac{x-3}{x^2+4x+3}$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3}{(x+2)(x+3)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3}{(x+2)(x+3)}$$

Worked Solutions

Rational Expressions

Q077 MCQ

A learner combines

 $\frac{2x-1}{(x-1)(x+2)} - \frac{x+4}{x+2}$. Correct the numerator line, then simplify the expression:

Solution

1. Keep one common denominator; never add the denominators.
2. Put the entire second numerator in parentheses before applying the sign.
3. After factoring and reducing, the correct result is $= -\frac{x^2+x-3}{(x-1)(x+2)}$.
4. Check the original denominator restrictions.

Correct option: B

Final answer

$$-\frac{x^2+x-3}{(x-1)(x+2)}$$

Worked Solutions

Rational Expressions

Q078 **Written****Simplify**

$$\frac{x^3 + 8x}{x^2 - x - 2} - \frac{8}{x - 2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x^2 + 2x + 4}{x + 1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x^2 + 2x + 4}{x + 1}$$

Worked Solutions

Rational Expressions

Q079 Challenge

Simplify the single-inner fraction together with the reciprocal-pair quotient; state the full domain.

Solution

1. Identify the two family triggers before manipulating the expression.
2. Simplify each component inside-out, keeping the original restrictions.
3. Combine the reduced components: $= -\frac{x^2 - 4x + 2}{2(x - 2)}$.
4. Audit every original inner and outer denominator.

Final answer

$$-\frac{x^2 - 4x + 2}{2(x - 2)}$$

Worked Solutions

Rational Expressions

Q080 Written

Simplify

$$\frac{1}{1 - \frac{1}{x-1}}.$$

Solution

1. The inner denominator requires $x-1 \neq 0$.
2. Multiply the outer numerator and denominator by $x-1$.
3. The result is $\frac{x-1}{x-2}$.
4. The outer denominator also excludes $x = 2$.

Final answer

$$\frac{x-1}{x-2}$$

Q081 Written**Simplify**

$$\frac{1}{1 - \frac{x}{x+1}}.$$

Solution

1. The inner denominator requires $x+1 \neq 0$.
2. Clear the inner denominator $x+1$ in the outer fraction.
3. The denominator becomes 1, so the result is $x + 1$.
4. Retain $x \neq -1$ from the original expression.

Final answer

$$x + 1$$

Worked Solutions

Rational Expressions

Q082 MCQ

Simplify

$$\frac{1}{1 - \frac{x+2}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 1$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

$$-x - 1$$

Q083 MCQ**Simplify**

$$\frac{1}{1 - \frac{x+3}{x+2}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 2$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$-x - 2$$

Worked Solutions

Rational Expressions

Q084 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+1}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+1}{x}$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

$$-\frac{x+1}{x}$$

Q085 MCQ

Simplify

$$\frac{1}{1 - \frac{2x + 3}{x + 1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x + 1}{x + 2}$.
4. State every restriction from the original denominators.

Correct option: C**Final answer**

$$-\frac{x + 1}{x + 2}$$

Worked Solutions

Rational Expressions

Q086 MCQ

Simplify

$$\frac{1}{1 - \frac{3x+1}{x+2}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+2}{2x-1}$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

$$-\frac{x+2}{2x-1}$$

Q087 MCQ

Simplify

$$\frac{1}{1 - \frac{x+4}{x+3}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 3$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$-x - 3$$

Worked Solutions

Rational Expressions

Q088 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+5}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+1}{x+4}$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

$$-\frac{x+1}{x+4}$$

Q089 Written**Simplify**

$$\frac{1}{1-x} + \frac{1}{1+x} \div \left(\frac{1}{1-x} - \frac{1}{1+x} \right).$$

Solution

1. Multiply both outer parts by $(1-x)(1+x)$.
2. The numerator pair becomes 2; the denominator pair becomes $2x$.
3. Reduce to $1/x$.
4. The original inner denominators and the outer denominator give $x \neq -1, 0, 1$.

Final answer

$$1/x$$

Worked Solutions

Rational Expressions

Q090 MCQ

Simplify

$$\frac{1}{x+1} + \frac{1}{x+4} \div \left(\frac{1}{x+1} - \frac{1}{x+4} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= \frac{2x+5}{3}$.
4. State every restriction from the original denominators.

Correct option: A

Final answer

$$\frac{2x+5}{3}$$

Q091 MCQ**Simplify**

$$\frac{1}{x+1} + \frac{1}{x+5} \div \left(\frac{1}{x+1} - \frac{1}{x+5} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= \frac{x+3}{2}$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$\frac{x+3}{2}$$

Worked Solutions

Rational Expressions

Q092 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2} \div \left(\frac{1}{x+2} - \frac{1}{x-2} \right).$$

Solution

1. Multiply numerator and denominator by $(x+2)(x-2)$.
2. The numerator pair becomes $2x$; the denominator pair becomes -4 .
3. The result is $-x/2$.
4. Keep $x \neq -2$ and $x \neq 2$.

Final answer

$$-\frac{x}{2}$$

Q093 MCQ**Simplify**

$$\frac{1}{x+1} + \frac{1}{x+3} \div \left(\frac{1}{x+1} + \frac{1}{x+3} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: C**Final answer**

1

Worked Solutions

Rational Expressions

Q094 MCQ

Simplify

$$\frac{1}{x+2} + \frac{1}{x+5} \div \left(\frac{1}{x+2} + \frac{1}{x+5} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: D

Final answer

1

Q095 MCQ

Simplify

$$\frac{1}{x+3} + \frac{1}{x+6} \div \left(\frac{1}{x+3} + \frac{1}{x+6} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: B**Final answer**

1

Worked Solutions

Rational Expressions

Q096 Written

Simplify

$$\frac{1}{x+y} + \frac{1}{x-y} \div \left(\frac{1}{x+y} - \frac{1}{x-y} \right).$$

Solution

1. Clear both inner denominators using $(x+y)(x-y)$.
2. The numerator pair becomes $2x$; the denominator pair becomes $-2y$.
3. Reduce to $-x/y$.
4. Keep $y \neq 0$ and $x \neq \pm y$.

Final answer

$$-\frac{x}{y}$$

Q097 Written**Simplify**

$$\frac{1}{1 - \frac{1}{1 - \frac{1}{x}}}.$$

Solution

1. Start with the innermost $1 - 1/x$.
2. Substitute that result into the next reciprocal layer.
3. The outer result reduces to $1 - x$.
4. The original denominators exclude $x = 0$ and $x = 1$.

Final answer

$$1 - x$$

Worked Solutions

Rational Expressions

Q098 Written

Simplify

$$\frac{1}{1 - \frac{1}{1 + \frac{1}{x}}}.$$

Solution

1. Simplify $1 + 1/x$ before entering the next layer.
2. Substitute the result into $1 - 1/(...)$.
3. The final result is $x + 1$.
4. Retain $x \neq 0$ and $x \neq -1$.

Final answer

$$x + 1$$

Q099 Written**Simplify**

$$\left(1 + \frac{x-y}{x+y}\right) \div \left(1 - \frac{x-y}{x+y}\right).$$

Solution

1. Let $u = (x - y)/(x + y)$, then combine $1+u$ and $1-u$.
2. Clear $x+y$ in the resulting numerator and denominator.
3. The reduced result is $\frac{x}{y}$.
4. Require $y \neq 0$ and $x + y \neq 0$.

Final answer

$$\frac{x}{y}$$

Worked Solutions

Rational Expressions

Q100 Written

Simplify

$$\left(x - \frac{2}{x-1}\right) \div \left(\frac{x}{x-1} - 2\right).$$

Solution

1. Combine the polynomial pieces over $x-1$ in the numerator and denominator.
2. Form the outer quotient only after both pieces are simplified.
3. Cancel complete factors to obtain $-(x+1)$.
4. The original denominators retain $x \neq 1$ and $x \neq 2$.

Final answer

$$-(x+1)$$

Q101 Written

Simplify

$$\left(\frac{x+1}{x-1} - \frac{x-1}{x+1}\right) \div \left(\frac{x+1}{x-1} + \frac{x-1}{x+1}\right).$$

Solution

1. Use $(x-1)(x+1)$ inside both the numerator and denominator.
2. The difference gives $4x$; the sum gives $2(x^2+1)$.
3. The simplified result is $\frac{2x}{x^2+1}$.
4. The original factors require $x \neq \pm 1$.

Final answer

$$\frac{2x}{x^2+1}$$

Worked Solutions

Rational Expressions

Q102 Written

Simplify

$$\frac{1}{1 + \frac{1}{1 + \frac{1}{1+x}}}.$$

Solution

1. Translate the three booster stages as an inside-out reciprocal chain.
2. Simplify the innermost stage $1+1/(1+x)$.
3. Continuing outward gives $\frac{x+2}{2x+3}$.
4. The stated model condition is $x>0$.

Final answer

$$\frac{x+2}{2x+3}$$

Quiz MCQ 1 Concept Quiz · MCQ

Simplify $\frac{x^2-4}{3x^2+7x+2}$:

Solution

1. Factor both polynomials and cancel $x+2$.

Correct option: A

Final answer

$$\frac{x-2}{3x+1}$$

Worked Solutions

Rational Expressions

Quiz MCQ 2 Concept Quiz · MCQ

Simplify

$$\frac{x+1}{x-2} + \frac{2}{x-2}:$$

Solution

1. The denominators match, so add the numerators only.

Correct option: A

Final answer

$$\frac{x+3}{x-2}$$

Quiz MCQ 3 Concept Quiz · MCQ

In

$\frac{3}{x^2+x} - \frac{3-x}{x^2+x}$, the numerator after the minus sign is... Which of the following is correct?:

Solution

1. Parentheses preserve the sign of every term in the second numerator.

Correct option: A

Final answer

$$3 - (3 - x)$$

Worked Solutions

Rational Expressions

Quiz MCQ 4 Concept Quiz · MCQ

Which statement is safe when simplifying a compound fraction?:

Solution

1. A common multiplier preserves the quotient and exposes the structure.

Correct option: A

Final answer

multiply top and bottom by the same inner denominator

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $\frac{x^2-9}{x^2-2x-3} \times \frac{x^2-4}{x^2+x-6}$ and state the original restrictions.

Solution

1. Factor all four polynomials and cancel common factors.
2. Final form: $= \frac{x+2}{x+1}$; retain original denominator exclusions.

Final answer

$$\frac{x+2}{x+1}$$

Worked Solutions

Rational Expressions

Quiz WRITTEN 6 Concept Quiz · Written

Simplify

$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}$ and state the original restriction.

Solution

1. Use the common denominator $x+1$ and combine the numerators.
2. Final form: $= x - 1$; $x \neq -1$ remains excluded.

Final answer

$$x - 1$$

Quiz WRITTEN 7 Concept Quiz · Written

Simplify

$\frac{x^2 - y^2}{x^2 - 2xy + y^2} - \frac{x + y}{x - y}$ in x and y .

Solution

1. Factor $x^2 - y^2$, use the common $(x - y)$ structure, and apply the minus sign to $x + y$.
2. Final form: $= 0$.

Final answer

$$0$$

Worked Solutions

Rational Expressions

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $(x^2 - 16)/(x - 4) = x + 4$ is not an identity for every real x .

Solution

1. The cancellation is valid only when $x - 4 \neq 0$.
2. At $x = 4$ the original expression is undefined, so the graph has a hole.

Final answer

$x \neq 4$; removable hole at $(4, 8)$

Answer key

Use the question number shown on each page.

Q001 A

Q002 B

Q003 C

Q004 D

Q005 A

Q006 B

$$Q007 \frac{4b^3}{3a^2}$$

$$Q008 \frac{3y^2}{10x}$$

$$Q009 \frac{3a}{4b^3}$$

$$Q010 \frac{1}{4ab^2}$$

Q011 B

Q012 A

Q013 A

Q014 B

Q015 A

Q016 B

Q017 C

Q018 D

Q019 A

Q020 B

Q021 C

$$Q022 \frac{3c^2}{a}$$

$$Q023 \frac{4a^2c^3}{b^2}$$

Q024 C

Q025 D

Q026 D

Q027 A

Q028 B

Q029 C

$$Q030 \frac{x-2}{3x+1}$$

$$Q031 \frac{2x-5}{x+2}$$

$$Q032 \frac{x}{x-3}$$

$$Q033 \frac{x+2}{x-4}$$

$$Q034 \frac{2(x-y)}{x(x+4y)}$$

$$Q035 \frac{x^2+x+1}{x+3}$$

$$Q036 \frac{x}{x-3}$$

$$Q037 \frac{x-4}{x(x-2)}$$

$$Q038 \frac{(x-5)(x+3)}{x(x+4)}$$

$$Q039 \frac{(x+3)(x+4)}{(2x-3)(3x+1)}$$

$$Q040 \frac{(x-3)(x+1)}{(x+2)(x+4)}$$

$$Q041 \frac{x-3}{(x-2)(2x-1)}$$

$$Q042 \frac{3ab^2}{4x^2}$$

$$Q043 \frac{(x-3)(2x-5)}{(x-1)(x+1)}$$

$$Q044 \frac{3}{x+3}$$

$$Q045 \frac{1}{x}$$

$$Q046 -\frac{1}{2} + \frac{15129\sqrt{123}}{2}$$

$$Q047 \frac{x^2+4x-2}{(x-2)(x+3)}$$

$$Q048 x-1$$

Q049 C

Q050 D

Q051 A

Q052 B

$$Q053 \frac{1}{x+1}$$

Q054 A

Q055 B

Q056 C

Q057 C

$$Q058 \frac{x+2}{x(x-1)}$$

$$Q059 \frac{x}{(x+1)(x+2)}$$

Q060 B

Q061 C

Q062 D

Q063 A

Q064 B

Q065 C

$$Q066 \frac{x+2}{x+1}$$

$$Q067 \frac{3x-2}{x(x-1)}$$

$$Q068 \frac{2x+15}{(x-1)(x+6)}$$

Q069 C

$$Q070 \frac{2x}{(x-2)(x+2)}$$

$$Q071 -\frac{2}{x(x-1)}$$

$$Q072 \frac{2}{(x+1)(x+2)}$$

$$Q073 \frac{2x^2-4x+3}{x(x-1)(x+1)}$$

$$Q074 \frac{3x-1}{x(x-1)}$$

$$Q075 \frac{2}{(x+2)(x+4)}$$

$$Q076 \frac{3}{(x+2)(x+3)}$$

Q077 B

$$Q078 \frac{x^2+2x+4}{x+1}$$

$$Q079 -\frac{x^2-4x+2}{2(x-2)}$$

$$Q080 \frac{x-1}{x-2}$$

Q081 $x+1$

Q082 D

Q083 A

Q084 B

Q085 C

Q086 D

Q087 A

Q088 B

Q089 $1/x$

Q090 A

Q091 A

Q092 $-\frac{x}{2}$

Q093 C

Q094 D

Q095 B

Q096 $-\frac{x}{y}$ Q097 $1 - x$ Q098 $x + 1$ Q099 $\frac{x}{y}$ Q100 $-(x + 1)$ Q101 $\frac{2x}{x^2 + 1}$ Q102 $\frac{x + 2}{2x + 3}$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{x + 2}{x + 1}$ Quiz WRITTEN 6 $x - 1$

Quiz WRITTEN 7 0

Quiz WRITTEN 8 $x \neq 4$; removable hole at (4, 8)